EXAMINATION QUESTIONS

QUESTION 1

SECTION 05

09.3.2014

To: micahelhart@ndgarchitects.com

From: candidate@ndgarchitects.com

CC:

Re: New 6 classroom extension to Christchurch school, Madchseter

Acting as Consultants to Roland Construction as part of a Design Build Bid

Dear Michael,

Thank you for your email…and sorry that you also have this dilemma to keep you up at night!

I have given your question some thought and would like to first set out my understanding of the issues:

- We have a good working relationship with Roland Construction built up over many years. This type of relationship is important to foster as it helps prevent the construction process from becoming adversarial and un-cooperative. The 3rd point of the RIBA Code of Conduct "Relationships" implies that relationships are important to maintain the standing of the architectural profession and of the projects that we are involved in with respect to the community at large.
- The prospect of more work is always appealing, particularly when it is with a contractor which who we are used to working with.
- The Board of Governors of the school has been a long term client that has developed a trusting relationship with our firm. They know and trust our level of competence and professionalism. Again, an understanding of the Code of "Relationships' and the RIBA Code of "Integrity" play a large part here.
- We are in the middle of a possible liability claim with regards to the recent accident at the school that has put into question our competence in the eyes of this client. However, we are dealing with the issue in a prompt and appropriately professional way following standards 10, 11, 8 of the ARB code where our professional duties are defined with respect to complaints, cooperation, and insurance requirements.
- You are concerned that our teaming up with Roland Construction to bid for this new extension will come off as having a "cavalier" attitude to the incident at the completion of the 4 classroom extension and our client relationship. Moreover, the client could perceive us taking part in the bid as a conflict of interest.

If we are to team up with Roland's I believe that we can ensure our professional integrity in the following ways:

- Per the RIBA Code; that we act honestly at all times. We should let our client know in writing well before hand that we have an opportunity to team up with this contractor to bid for this new work. It may be the case that they will appreciate the fact that the architect consultant in the contractor team will be a known party.
- Per the 3rd Standard of the ARB code; we should promote our professional services in a truthful and responsible manner. This would done by ensuring that at all time our scope of work and involvement in this bid is fully understood by the Board of Governors and that our actions with regards to the recent accident at the school would have no bearing on the new contract we would be involved in.
- In the same vein, Roland Construction would have to have a clear understanding of our current relationship with the Board of Governors (a fact that I am sure it not lost on them as they know it will help support their bid) but also that any future development of issues regarding the accident at the school will not have any bearing on our relationship with them as their architect consultants for the Design Build contract.
- I believe that it would be good practice to suggest a meeting between ourselves, the contactor and a representative of the Board of Governors to lay all the relevant issues out on the table regarding this bid so that each of the parties can air its point of view and resolve any concerns about integrity, conflicts of interest and working relationships.

I trust these thoughts will be useful to you

Best regards

NDG ARCHITECTS
Wharf Mill Paccadilly, Madchester, MAD PPA

EXAMINATION QUESTIONS

QUESTION 2

SECTION 05

09.3.2014

To: BelindaCarter@ndgarchitects.com

From: candidate@ndgarchitects.com

CC:

Re: Community Performing Arts centre

Your presentation – "Stakeholders"

Dear Belinda,

Thank you or your email regarding your upcoming presentation. I am delighted to help you with some thoughts on the topic of "Stakeholders" for the Performing Arts Centre. A prestigious and high profile project such as this is likely to have many parties that are interested or affected by the project:

- Stakeholders can be divided into Direct Stakeholders and Indirect Stakeholders
- Some are passive, some are active and all have varying degrees of influence
- In the same way some stakeholders may initiate and drive projects forward, others are more obstructive
- Analysing the position and interest of multiple stakeholders is essential for properly informed brief and successful project.
- Stakeholders will have varying degrees of experience with construction projects so the process must be managed carefully in order to be effective
- Therefore suggest the formation of a core group of "Key Design Team" members who all should have at least some experience of the construction process. In this way the strategic and briefing effort can be efficiently targeted when need be and at other times information can be more widely disseminated
- Key Design Team - Project Executive (School Governors and their advisors; including finance and property managers); Design Advisory Executive (Architect and design professionals), User Group Executive (Users, Operations and Specialists).

Key stakeholders:

- School Governors
- Education and Art department's of Local Authority
- Local and National Performing Arts Organisations
- Client appointed advisors
- Architect and other consultants – Landscape, Engineers, Fire, Inclusive design

Direct stakeholders:

- Teaching and support staff
- Professional users
- Students and amateur users
- Operations and maintenance
- Security consultants
- Specialist consultants and suppliers such as acousticians, sound engineers
- LPA – Planning and Building control
- Cost consultants
- Heath and safety
- Donors and local sponsors
- Banks and funders
- Central Government funding bodies

Indirect stakeholders:

- Municipal Service providers
- Highways and transport
- Local community groups

With regards to how the project and brief will shape up, you suggest to refer to the first 3 stages of the Plan of Work and show how the stakeholder's interests are accommodated into those stages:

RIBA Stage 0 – Strategic definition

Identification of Client's needs and objectives, business case and any possible constraints on development. Feasibility studies and assessment of options.

  - Strategic Kick off meeting and a subsequent workshop will be held with Key Design Team to ensure business case and strategy has been properly considered – new build is correct choice etc.

- Purpose of workshop will be to "set the scene" upon which the development of the brief will be based
- Determined necessary services and feasibility within site constraints
- Identification of procurement route options, procedures, organisational structure and range of consultants that may need to be engaged – CDM consultant etc.
- Develop the project Mission Statement
- Findings will be reported back to the stakeholder body as a Statement of Need which will form the basis of the Design Brief
- Statement of what is expected in response to this Statement
- State the sorts of decisions that are needed and from whom
- Collation of responses, review and analysis with Key design Team
- Timeframe 3 to 4 weeks

RIIBA Stage 1 – Preparation and Brief

- Design brief kick off meeting with Key Design Team to establish a working framework between parties
- Development of a project directory and Organization chart
- Site Survey and environmental studies – existing conditions, services and highways
- Accessibility assessments to and from the site
- Thoroughly review potential space program, planning "bubble" diagrams and related project information such as planning restrictions and the relationship to the LPA development plan. Preliminary approach to planning department.
- Review position of stakeholders with regards to project risk. Different stakeholders will have different attitudes to risk, whether it is in the building procurement route or as an end user
- Conduct "visioning" workshops (at least 2) to verify project goals and objectives, gather and analyse information, identify programmatic strategies, establish quantitative requirements; establish maintenance, inclusive design principles as well as service and security requirements, quality standards, building life expectancy, heath and safety policy and, finally, develop the design criteria.
- Design advisory executive will lead the workshops with an intent to establish and document the planning and design aesthetic desired by School Governors in support of the Madchester Academy mission and goals
- Project budget will be developed in consultation with Key Design Team members as well as Cost Consultant executives
- Developed Project Brief document will then be shared with the entire Stakeholder group when stakeholders will have the opportunity review and ask questions and make comments
- Review and analysis of comments and findings at Stakeholder group meeting
- Timeframe 5 weeks

RIBA Stage 2 – Concept design

- Develop the general design direction for the building in support of the brief.
- Coordinate the design direction with the development of functional floor plans
- Conceptualise specialty improvements to the plan that may include specialised flooring systems, specialised audio systems and lighting, technology integration, environmental graphic concepts, the integration of art, sustainability ideas and any other concepts that make the design of the Performing Arts centre unique to the Academy's mission and goals.
- Review the concept design with the Key design Team members to make sure that all stakeholders interests are being met
- Take comments from the review and make necessary adjustments and changes
- Review concept with the Local Planning Authority so that issues can be raised early and so that eventually the LPA may take some "ownership" of the design to help get it approved.
- Review concept with Stakeholder group in the form of a design presentation – drawings, renderings etc.
- Format of questions and comments to be managed so that feedback is clear
- Timeframe 6 to 8 weeks

RIBA stage 3 – Developed design

- Concept Design is further developed taking into account stakeholders comments as well as input from LPA
- Design workshops with Key Design Team in the development of requirements for individual spaces and functions
- Design professionals will manage Coordination with proposed structural and building systems
- Design professionals will gather cost information and make sure it is aligned to the project budget with project cost consultant
- Review of developed design with Key design team members
- Review of developed design with invited members of the larger Stakeholder group though a presentation with an invitation to give formal comments - site layout, planning, elevational treatments, construction and environmental systems.
- Review of comments by Key Design team and decision if any should be implemented/
- Review of developed design with LPA and other statutory bodies such as highways and municipal service providers. Preparation of final submission documents for Planning approval
- Timeframe 8 to 10 weeks

EXAMINATION QUESTIONS

QUESTION 3

SECTION 05

09.3.2014

To: micahelhart@ndgarchitects.com

From: candidate@ndgarchitects.com

CC:

Re: 4 Classroom refurbishment

Door injury incident

Dear Michael,

Thank you for filling me in in detail on this situation. I just heard about it in the office when I can back in today and it certainly has me very concerned too. With regards to your questions, I have prepared the following comments for your meeting with the directors.

1 You are correct that we specified finger guards for all doors for the 4 classroom refurbishment. This was an attempt to cover all eventualities for the current and future use of the classrooms. As you know we were all under a lot of pressure to handover the project, not least pressure applied by the head teacher so that she may take possession of the classrooms. I realize that issuing the Certificate of Practical Completion has a major impact on the contract; not least that it releases the contractor from his insurance obligations. I had a discussion with the contractor regarding the time needed to fit the specified finger guards. My assessment of the situation was that the doors concerned could be considered low risk from the point of finger trapping (they are of low weight and have self closers with arresting action) and, from my understanding at the time, there was a the relatively low risk that young children would be using the classroom in the first couple days of handover, I decided to issue the Certificate of Practical Completion whilst noting carefully that one of the outstanding items was the installation of the finger guards. I also pointed out to the contractor that these items in particular should be installed as soon as possible.

2 I think that we should make sure that we have clear evidence of our intention that the finger guards should have been provided in time for the completion of the project. That they are specified for every door should help our case demonstrating our competency in design and that we have been following regulation 11 of the CDM regulations not just with regards to on site operations but of the health and safety of the building's users and operators. We should inform our Insurance Company of a potential claim of negligence. If we receive a claim of negligence from our client or a third party it will be important to admit nothing but instead work closely with our Insurance company to make sure that they understand the situation fully. We may also need to seek legal advice from a qualified lawyer

3 I have checked into this issue in the office copy of the Architect's Legal Handbook. "The architect owes his standard duty of care in tort to not have injury caused as a result of his negligence" p 324. However in this case I do not think I issued the CPC negligently. The provision of figure guards is not mandatory for a public building, not even a school. It is our dedication to best practice that made us decide to specify them in the first place I believe that this illustrates that we have approached this matter with a "standard duty of care". On their website, The Heath and Safely executive suggests figure guards on doors in schools should be provided based upon an assessment of the risk of individual doors so I do not think there is case for negligence there, rather a series of unforeseen and unfortunate events that led to this accident.

4 This leads me to consider the role of the CDM consultant in this matter. As you know we have had a good working relationship thought the project. Our assessment of the risk of finger traps on the doors in the project was deemed low when we were putting together the specifications for the project. Although it would have been best practice to have the fully updated health and safety file ready for practical completion this had no real bearing on this incident. The CDM consultant was in receipt of my notes that listed the installation of the finger guards as outstanding so I believe that we did our best to keep him informed of items that could change the project Health and Safety Plan.

5 In hindsight I would have organized a more formal project handover meeting at Practical Completion and made sure that the head teacher was clear on what items were still outstanding and have her sign off on any items that posed a safety risk, however slight. In that way she could have organised additional supervision or flatly not allowed children below a certain age to use the classroom. In addition, instated of making a general decision to allow the contractor install the fingers guards past CPC, I could have assessed which doors posed the higher risk of finger trap incidents and had the contractor prioritise those in his schedule to fix the outstanding items. A programme of dates of when these items would be rectified would have been a useful addition. I also could have informed our Insurance Company of the schedule of outstanding items and the programme.

Please find attached a draft letter per your request. Please let me know if you require any further assistance.

Candidate

NDG ARCHITECTS
Wharf Mill Paccadilly, Madchester, MAD PPA

NDG ARCHITECTS
Wharf Mill
Paccadilly
Madchester
MAD PPA

DRAFT
September ## 2014

Christchurch School
Madchester

For the attention of: Ms. Janet Jones, Head Teacher

RE: Recent Accident in new Classroom refurbishment

Dear Ms. Jones,

I am writing to follow up on your phone call yesterday regarding the accident to the school. Our practice takes any incident where an injury occurs on any one of our projects, whether under construction or completed, very seriously. We trust that the child that is injured is doing well under the circumstances.

To that end we have launched an full investigation of the facts here in the office. We will also consult with the Construction Design and Management Consultant that worked on the project. You told me that you have spoken to her and you were concerned that you were told that the contractor's updated Heath and Safety plan would be available only in a couple of weeks. While it is not best practice for the contractor to have this amount of delay the presence of the file is not critical to this issue and the file has been kept up to date during the whole construction process according to the statutory regulations.

Once we have completed our investigations into the circumstances this incidence occurred we would like to have the opportunity to review the incident and the surrounding circumstances with you and the Board of Governors so that you may get a balanced picture. We expect the investigation to take no more that three or four days after which I would like to contact you again to arrange this meeting.

Once again, we express our sincere concern about this regrettable incident and look forward to meeting to help answer any questions that you or the Board of Governors may have.

Yours sincerely,

Mr. Sven Berg

Architect
On behalf of NGD Architects

EXAMINATION QUESTIONS

QUESTION 4

SECTION 05

09.3.2014

To: micahelhart@ndgarchitects.com

From: candidate@ndgarchitects.com

CC:

Re: Classroom extension at Christchurch School Madchester

Draft Fee Proposal for Consulting Services Stages 5 through 7

Dear Michael,

Please find below my draft of the scope of services and fee calculation per your email request.

Dear Mr Henratty,

I am very pleased that the preparation of your bid for the Design Build Contract for the 6 classroom extension is progressing and happy that we are partnering with your firm once again. Now that we have agreed on a fee for basic working drawings we have been considering what other consulting tasks that we can be involved in to ensure the successful completion of this project and give The Board of School Governors the assurance that everything will be duly taken care of. Of course, Mobilisation, construction and handover are primarily Roland Construction's responsibility and we will not be acting as contract administrators but rather as your architectural Consultants. I thought it would be helpful to review what aspects of the project our firm may or may not be involved in a consultancy role to avoid any possible misunderstandings or disputes. I have broken our scope recommendations into subheadings with regards to the different stages and processes with a fee for the services at the end.

RIBA Stage 4 – Technical design (Additional Services)

I assume that the terms of our previous fee proposal for this stage is still current and is included in your bid documents. In this phase although we will be your consultants, we will be working as lead designers and coordinating the other consultant members of the design team. I assume that you made clear any conditions in respect of the method of structuring and supply of production information when we submitted our fee proposals for the working drawings. Some additional points to consider with regards to our agreement:

- We will check with the client that they have appointed an Employers agent to act on their behalf. It will be the employer's responsibility to confirm in writing acceptance of proposals and supplied information with regards to the Employers Requirements. We will check to make sure that the information provided is adequate and generate a query list for any items that require clarification or if there any discrepancies or omissions. This will be important to avoid any problems later down the line.
- We will assist you with ensuring the requirements of any Design Submission procedures are met
- We will check that the client is aware of their responsibilities under the CDM legislation and that they have appointed a CDM Consultant. We will assist you in the preparation of your "Construction Phase Plan" so that it complies with the regulations. We will assist you in developing method statements
- At the end of this stage we will need to submit an application for Building Regulations approval which was also not included in our original fee proposal. We suggest a Full Plans submission.
- There may be other discussion and approvals needed from bodies such as highways for access to the site with regards to materials off loading which we will assist with.
- We will work with you, your cost estimator or indeed a client appointed quantity surveyor to cost the project against the client agreed construction budget and make value engineering suggestions and changes to the design within a reasonable level.

Stage 5 - Construction

As your consultants during the Mobilisation phase (estimated time frame 4 weeks) we will:

- Work with you to ensure Contract documents are properly prepared – we suggest JCT DB 2011to be used
- Agree on procedures for the exchange of information between you and other consultants and project stakeholders
- Work with you to ensure the prosper contractor insurances are in place although maintenance of this insurance will be your ultimate responsibility
- Arrange and manage an appropriate agenda for formal project Pre-start meeting. At this meeting we can confirm our role in relation to the rest of the consultant team
- Consult with the Employers Agent to ensure that any insurance obligations on the employers side are met
- Consult with the employers Agent and yourselves to check the scope of professional consultant services that have been agreed for the construction stage of the project
- Check that all notices granting planning permission and approval under building regulations are to hand and that approvals are still within valid time limits
- Check with the CDM Consultant that the HSE has been given particulars required by law and that your Health and safely plan has been approved in writing

- Work with you on site planning and accommodation taking into account the various site specific conditions and the fact that we are working on a public occupied building.

During the Construction Stage (estimated Timeframe 20 weeks), we will not be acting as Contract Administrators but as consultants offering the following assistance:

- Check that all coordinated information required is available to the construction team
- Assist in the management of the Contractors Master Programme although we will not be assuming any responsibly that it is adhered to
- Assist in the distribution and updating of your Health and Safety plan while you hold ultimate responsibility for it
- Confirm scope and frequency of site meetings in addition to visiting the site as designers to observe if work is conforming to the design drawings and carry out quality control observations on your behalf. We will not be assuming any roles on site that are the duties of the Principle contractor or indeed in the administration of the contract
- Review design information from subcontractors and provide reasonable further information if necessary. We do not plan to attend every one of your production meetings with your subcontractors
- Assist with your Review of compliance with the statutory regulations and contract requirements
- Assist with the setting up of procedures for issuance of certificates of payment. We will not be issuing certificates of payment
- Assist in the keeping the Employers agent informed as to the progress of the job, any material changes in the design due to unforeseen conditions and, of course, changes to the programme, cost or quality expected by the employer. We will not be consulting on costs or reporting costs or changes to the Employers agent
- Be available to give general advice on operation as the works are completed considering any possible extensions of time claims and future loss and expense claims but we will not be issuing any of these claims to the Employer
- We will liaise with the employers agent to determine if there are any required Practical Completion procedures
- We will assist your Site Agent in the preparation of contractor quality control and snagging lists

Stage 6 – Handover and Closeout:

During this stage we propose the following consultant services:

- We will assist with organising final commissioning of the project including making sure that maintenance contracts are in place

- Assist with the arrangement of a formal handover meeting and advise on issues relating to Practical completion
- We will prepare “as built” drawings
- We will assist in the preparation of the final Health and Safely file
- We can assist in the preparation of a building owner maintenance manual which is probably very useful for a school

Stage 7 – In use (Optional)

As the employer is a client body that both you and our firm are interested in retaining for future work, we think that it would be good to suggest a post occupation evaluation as an additional service. Upon agreement of this phase from the employer we will:

- Arrange interviews with key personnel from the client team
- Conduct an appraisal of the feedback from the client team and hold a debriefing exercise with the other project consultants
- The findings of this post occupancy evaluation can be formatted and presented to the employer in collaboration with you.

One import point of our proposed consultant relationship to you is with regards to design obligations. An architect’s normal obligation, like that of any other professional, is to use reasonable skill and care. In contrast, the normal design and build liability for a contractor is to produce an end result which is “fit for it’s intended purpose”. The final terms of our contact should be worded to respect this difference, as “fitness for purpose” liability will not be covered by our PII insurance.

With regards to the above scope of services we are pleased to submit the following fee proposal as a lump sum fee of £ 17,610

The fee will be billed monthly on percentage of completion basis in the following increments:

Stage 4 Technical design (Additional Services)	£ 2,100.00
Stage 5 Construction	£ 13,600.00
Stage 6 Handover	£ 790.00
Stage 7 In Use (Optional)	£ 1120.00
TOTAL	£ 17,610

PROJECT BUDGET SHEET

PROJECT: Classroom Extension Christchurch School

NDG Architects Ltd.

Stage	Estimate
Stage 0/1	Est. ## Weeks
Stage 2	Est. ## Weeks
Stage 3	Est. ## Weeks
Stage 4 (Additional Services)	Est. 4 Weeks
Stage 5	Est. 24 Weeks
Stage 6	Est. 1 Week
Stage 7	Est. 2 Weeks

Calculation of Hourly Charge Out Rates

	salary	NI %	OH	NI Amt	OH Amt	Sum	20% Profit	Sum with Profit	Base Billable Hrs	Hrly Rate	Input
director	52,000	0.13	1.00	6,760	52,000	110,760	22,152	132,912	1,816	73	73
associate	38,000	0.13	1.00	4,940	38,000	80,940	16,188	97,128	1,816	53	53
architect I	35,000	0.13	1.00	4,550	35,000	74,550	14,910	89,460	1,816	49	49
architect II	31,000	0.13	1.00	4,030	31,000	66,030	13,206	79,236	1,816	44	44
architect tech 1	27,000	0.13	1.00	3,510	27,000	57,510	11,502	69,012	1,816	38	38
architectural assistant	24,000	0.13	1.00	3,120	24,000	51,120	10,224	61,344	1,816	34	34
architectural tech 2	21,000	0.13	1.00	2,730	21,000	44,730	8,946	53,676	1,816	30	30

	4 weeks	Stage 4	4 weeks	4 weeks	4 weeks	4 weeks	4 weeks	4 weeks	Stage 5	1 week	Stage 6	2 weeks	Stage 7	TOTAL HOURS & FEE
director	0	0	0	0	0	0	0	0	0	0	0	0	0	0
associate	4	4	4	4	4	4	4	4	24	1	1	1	1	30
architect I	16	16	32	25	25	25	25	32	164	4	4	8	8	192
architect II	0	0	0	0	0	0	0	0	0	0	0	0	0	0
architect tech 1	0	0	0	0	0	0	0	0	0	0	0	0	0	0
architectural assistant	32	32	50	32	32	32	32	32	128	16	16	20	20	196
architectural tech 2	0	0	0	0	0	0	0	0	0	0	0	0	0	0
TOTALS	52	52	86	61	61	61	61	68	316	21	21	29	29	418
% complete hrs		12%							76%		5%		7%	100%
		2,083							13,686		791		1,123	17,683.69

227 working days per year exluding hoildays etc. working an 8 hour day

EXAMINATION QUESTIONS

QUESTION 5

09.3.2014

To: SvenBerg@ndgarchitects.com

From: candidate@ndgarchitects.com

CC: micahelhart@ndgarchitects.com

Re: Richard III College

Water Booster tank Incident

Dear Sven,

I am sorry to hear about this situation. I am happy to help you with dealing with the incident and its likely repercussions.

Firstly, with regards to responsibility I have the following thoughts:

We have worked with the contractor before and have not had cause to doubt his competence before. The contractor's hasty reaction to clearing the site of resultant mess can be seen a testament to his good workmanship record as well as the high standard of work that has been performed on this project to date.

The failure of the tank could be due either to be some sort of defect in its manufacture or it was specified incorrectly or it was installed incorrectly. To determine If the tank was not designed to be situated in a roof or there was not adequate or correct structure supporting it we will have to review the Mechanical engineer's drawings and specifications as well as the plumbing subcontractors shop drawings and submittals to make sure that they comply with the Engineering drawings and specifications. We will need to consult with the project engineer in order to review this issue effectively. We will also need the engineer to consult directly with the tank manufacturer to check their record of this particular tank to ensure that it met their quality control standards.

The contractor has expressed his concern to you that the tank may not, after all, have been installed properly. It will be important for him to bring all the relevant paperwork regarding the ordering and installation of this tank up to speed so that the situation can be assessed with the proper information.

The client raises a point about having a waterproof floor in the tank room; we should double check with our engineer if this is fact a requirement – I could not find anything in the building regulations other than that appliances such as these should be installed in workmanlike manner and all directions should be properly followed - Approved Document 7. The addition of a drip pan under the tank would have helped with the mitigation of any potential small leaks from the pipe connections to the tank but would not have helped a catastrophic flood such as the one our client experienced. Additionally we will want to discuss with the engineer his decision to place the water tank over the existing library and if any alternative locations were or are available now that the contractor will need to replace the tank. If the engineer has made a mistake in the specifications then he will need to refer to his PII insurance company for advice on how to proceed.

If it is found that that the tank was correctly specified and was installed according to the correct drawings and specification in a good workmanlike manner it may be necessary to employ the services of third party mechanical engineer or a specialist Forensic mechanical engineer to prepare a report gathering the available evidence on site and in the technical documents.

As the contract administrators we should monitor the process to ensure that the correct investigatory path is taken to determining what party or elements is at fault. That it was the contractor's sons company has no bearing on the case as the ultimate responsibility for the sub-contractors works rests with the general contractor although he may certainly seek damages from one of his subcontractors or suppliers.

With regards to the works progressing and next steps, I have looked back at the JCT SBC 2011 Contract Conditions and have the following points to make:

There has been damage to new works and existing finishes. The Contract includes provisions and options of insurance to indemnify the employer in respect of losses for damage to new works and existing buildings. This is referred to in section 6. Schedule 3 lists the different forms of insurance that are taken out on project. As this project involves work that is carried out to an existing building two insurances, both taken out by the employer will be in effect – Option C in the Insurance of Works particulars. The existing structure and contractor must be insured against "Specified Perils" as defined in Clause 6.8. New works in existing buildings must be covered by an "All Risks" insurance policy. These policies will be "Joint Names Policies naming the contractor and the employer and include coverage for "Specified Perils" as noted in the contract – these include the "escape of water from any water tank, apparatus or pipe".

Now that contractor has informed us, the Contract Administrators, of the damage it is imperative that the insurers are immediately informed. The employer also retains the right to terminate the employment of the contractor if they can provide "just and equitable" cause such as proof of negligence. All monies due under the insurance policy are paid direct to the employer.

Although, of course, this incident has caused a major disruption on site the works should proceed where possible. Any certificates of payment that have already been issued are not affected by the occurrence of the damage and should be paid. Any work that was completed after the most recent interim certificate but then subsequently damaged should also be included in the next interim certificate of payment.

In addition, under Clause 2.29, the contractor is entitled to an extension of time for delay caused by loss or damage due to one or more of the "Specified perils" as well as for claims for loss and/or expense.

I would advise that we work with the contractor to assess the best way to cordon off the damaged areas of the project and try to proceed with other works so as minimize the delay to the project programme.

I hope that these notes are useful to you. In addition as you requested, I have drafted a response to the client outlining the current contractual position.

Best regards,

Candidate

NDG ARCHITECTS
Wharf Mill Paccadilly, Madchester, MAD PPA

NDG ARCHITECTS
Wharf Mill
Paccadilly
Madchester
MAD PPA

DRAFT
September ## 2014

Richard III College
St Peters Way
Roachdale
OL3 2RT

For the attention of: Mr Earl Rivers

RE: Recent Water Damage to the Library

Dear Mr Rivers

We are in receipt of your letter regarding the above damage and have been in contact with the Contractor who informed us late last night of the occurrence. Of course, we appreciate your concern as to what exactly happened and who is at fault and also that school year is about to begin but we, as the contract administrators, have a duty to also explain to you the right procedures and the current contractual situation.

Firstly, you need to immediately inform your insurance provider about the incident. We will work with you, your insurance company and the rest of the project team to check if the works were specified and executed properly and if any of the installed equipment may have been faulty. The Joint Names Policy that the College, as the Employer in the contract, took out to cover both the new works and the existing building covers the damage that may be caused by a, so called, “Specified Peril” such as the water tank leaking water.

Secondly, it should be understood that the terms of the contract do not guarantee that that College will be able to claim liquidated damages from the contractor. As the damaged was caused by a “Specified Peril” the contractor is entitled to claim for an extension of time for delay and loss and/or expense. However, it should be recognized that the contractor has been cooperative in immediately helping to the clear up the mess and has performed all the works so far in a high quality workmanlike way and has made no claims as of yet.

We would like to suggest that it is therefore in all our interests to continue to work together to resolve this circumstance without resorting to premature claims against the contractor. In addition with regards to your enquiry as to the location of the tank and if any provision for a water proof floor should have been made, we will review these questions with the project engineer and get back to you with his answer.

Additionally, we are aware that the school year is about to begin and the works are nearing completion. We propose to work with the contractor to devise an appropriate way to partition the works from the rest of the school so that the balance of the work can continue. All current insurances continue to be in place while Practical Completion is not certified so you can rest assured that we are all doing our utmost the keep the project on track while we work together with the relevant parties to resolve this unfortunate occurrence.

Yours sincerely,

Mr. Michael Hart

Associate Director
On behalf of NGD Architects

EXAMINATION QUESTIONS

QUESTION 6

SECTION 05

09.4.2014

To: BelindaCarter@ndgarchitects.com

From: candidate@ndgarchitects.com

CC:

Re: Laanders House Refurbishment

Stove Issue

Dear Belinda,

I have read your email and have the following thoughts for your consideration.

Key issues:

1. The firm referred to Approved Document J of the building regulations in the design of the fireplace, flue and associated works for this decorative fireplace. The surround was designed with our normal "reasonable duty of care" following the regulations and best practice
2. The full plans Building Application was approved and the works were inspected by the local authority and awarded a Completion Certificate
3. The stove, although wood burning, was primarily a decorative item and not producing significant heat output
4. The stove was an owner supplier item and was installed at the owners risk by a subcontractor hired by himself, outside of the IC 11 building contract
5. The installer, in acting directly for his client, Mr Landers, is wrong to claim that the flue is too short or that hearth and proper ventilation was missing at this stage. If the installer had made the installation in a proper workmanlike way he should have pointed these defects out to the contract administrator and the general contractor at the time of his installation.
6. His claim regarding the flue, hearth and ventilation may be simply spurious, as we have issued the Final Certificate having checked that the works included in the IC 11 building contact were performed and completed in the proper manner. The flue, heath and ventilation would have all been work done under the building contract
7. If the flue, heath and ventilation defects are actually present then the Building Control officer who issued the Completion certificate may be at fault for missing the deficiencies in the installation as it seems to me, at this time, that the design of the fireplace and the stove were included in the Building regulations application.
8. We will have to check the site to see if the flue, hearth and ventilation elements were correctly installed by the general contractor and we have not issued the Final Certificate in error.
9. Additionally we should check our plans and specifications to make sure that they were indeed correct.

10. During this site visit we can also determine if in fact the stove is somehow faulty or indeed incorrectly installed by the ex-contact specialist or he failed to point out defects to the general contractors installation of the flue, heath and ventilation requirements

Conclusion:

It appears that the practice could be at fault if the works were indeed deficient and we issued the Final Certificate in error. However the Completion Certificate issued by Building control seems to support the case that the works were correctly completed by the general contractor – provided that the Inspector did not make an error. It would therefore appear that the specialist ex-contact installer may be at fault or there is some sort of fault with the stove itself.

I hope that the above points are useful to you and I look forward to discussing this matter further after we have gained a clearer understanding of the site conditions and have reviewed our drawings.

Best regards,

Candidate

NDG ARCHITECTS
Wharf Mill Paccadilly, Madchester, MAD PPA

QUESTION 7

09.4.2014

To: micahelhart@ndgarchitects.com

From: candidate@ndgarchitects.com

CC:

Re: Oldsworth barn conversion

Draft Plan of work

Dear Michael,
Thank you for your email. Yes, I attended a very instructive lecture about the 2013 plan of work and it has been a very interesting exercise to use the online tools to develop a:

- Draft Plan of Work
- Project Roles table
- Contractual Chart
- Schedule of services for Stage 0 and Stage 1 – you can refer to the sections pertaining to our firm to see the services that I have suggested we provide and project lead and project architects.

Please find them attached.

Additional Services

Considering the nature of this project and its location I believe the traditional procurement route will be the best way to proceed. Certainly the clients desire to create a sustainable house I have noted the inclusion of a Sustainability Consultant within the project team to help advise on strategies, environmental impact reduction and construction and refurbishment methodologies. I know that Sue Allen (Associate Director) is an accredited Sustainability Expert so I thought that perhaps we could offer her services in the role of Sustainability Consultant as an additional service our office could provide. Additionally, as you know I have quite extensive experience at my previous employment in Interior Design. I would suggest that the firm could also offer Interior Design services as an additional service and I would be happy to work with you and the rest of the team in developing interior design concepts for finishes and furniture for this client. In this way we can also ensure that sustainable concepts are carried on through all the way into the finishing out of the barn conversion too. I believe that this would be of added value to the client.

I look forward to hearing your thoughts

RIBA Plan of Work 2013

This **Project** version of the RIBA Plan of Work 2013 has been prepared by **ndg architects ltd** for use on its project **Oldsworth Barn Conversion**. It has been prepared on the basis of a **Traditional Contract** procurement route. The RIBA Plan of Work 2013 organises the process of briefing, designing, constructing, maintaining, operating and using building projects into a number of key stages. The content of stages may vary or overlap to suit specific project requirements. The RIBA Plan of Work 2013 should be used solely as guidance for the preparation of detailed professional services contracts and building contracts.

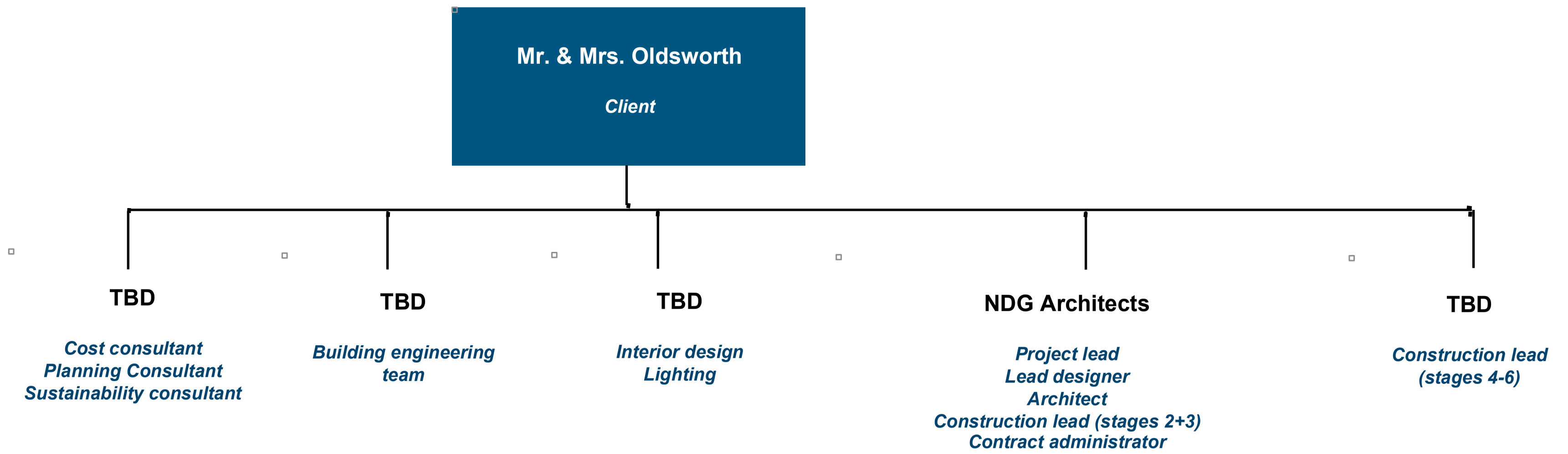
www.ribaplanofwork.com

Tasks ↓ / Work Stages →	0 Strategic Definition	1 Preparation and Brief	2 Concept Design	3 Developed Design	4 Technical Design	5 Construction	6 Handover and Close Out	7 In Use
Core Objectives	Identify client's **Business Case** and **Strategic Brief** and other core project requirements.	Develop **Project Objectives**, including **Quality Objectives** and **Project Outcomes**, **Sustainability Aspirations**, **Project Budget**, other parameters or constraints and develop **Initial Project Brief**. Undertake **Feasibility Studies** and review of **Site Information**.	Prepare Concept Design, including outline proposals for structural design, building services systems, outline specifications and preliminary **Cost Information** along with relevant **Project Strategies** in accordance with **Design Programme**. Agree alterations to brief and issue **Final Project Brief**.	Prepare Developed Design, including coordinated and updated proposals for structural design, building services systems, outline specifications, **Cost Information** and **Project Strategies** in accordance with **Design Programme**.	Prepare Technical Design in accordance with **Design Responsibility Matrix** and **Project Strategies** to include all architectural, structural and building services information, specialist subcontractor design and specifications, in accordance with **Design Programme**.	Offsite manufacturing and onsite **Construction** in accordance with **Construction Programme** and resolution of **Design Queries** from site as they arise.	Handover of building and conclusion of **Building Contract**.	Undertake In Use services in accordance with **Schedule of Services**.
Procurement *Variable Task Bar	Initial considerations for assembling the project team.	Prepare **Project Roles Table** and **Contractual Tree** and continue assembling the project team.			Design Team Stage 4 output issued for tender. Tenders assessed and **Building Contract** awarded. Specialist contractor Stage 4 information reviewed post award.	Administration of **Building Contract**, including regular site inspections and review of progress.	Conclude administration of **Building Contract**.	
Programme *Variable Task Bar	Establish **Project Programme**.	Review **Project Programme**.	Review **Project Programme**.	Review **Project Programme**.	Specialist subcontractor design work undertaken in parallel with Stage 5 in accordance with **Design** and **Construction Programmes**.			
(Town) Planning *Variable Task Bar	Pre-application discussions.	Pre-application discussions.	Pre-application discussions.	Planning application made at end of stage using Stage 3 output.	Planning conditions reviewed following granting of consent and, where possible, concluded prior to starting on site.			
Suggested Key Support Tasks	Review **Feedback** from previous projects.	Prepare **Handover Strategy** and **Risk Assessments**. Agree **Schedule of Services**, **Design Responsibility Matrix** and **Information Exchanges** and prepare **Project Execution Plan** including **Technology** and **Communication Strategies** and consideration of **Common Standards** to be used.	Prepare **Sustainability Strategy**, **Maintenance and Operational Strategy** and review **Handover Strategy** and **Risk Assessments**. Undertake third party consultations as required and any **Research and Development** aspects. Review and update **Project Execution Plan**. Consider **Construction Strategy**, including offsite fabrication, and develop **Health and Safety Strategy**.	Review and update **Sustainability Strategy**, **Maintenance and Operational** and **Handover Strategies** and **Risk Assessments**. Undertake third party consultations as required and conclude **Research and Development** aspects. Review and update **Project Execution Plan**, including **Change Control Procedures**. Review and update **Construction** and **Health and Safety Strategies**.	Review and update **Sustainability**, **Maintenance and Operational** and **Handover Strategies** and **Risk Assessments**. Prepare and submit Building Regulations submission and any other third party submissions requiring consent. Review and update **Project Execution Plan**. Review **Construction Strategy**, including sequencing, and update **Health and Safety Strategy**.	Review and update **Sustainability Strategy** and implement **Handover Strategy**, including agreement of information required for commissioning, training, handover, asset management, future monitoring and maintenance and ongoing compilation of **'As Constructed' Information**. Update **Construction Strategy** and **Health and Safety Strategies**.	Carry out activities listed in **Handover Strategy** including **Feedback** for use during the future life of the building or on future projects. Updating of **Project Information** as required.	Conclude activities listed in **Handover Strategy** including **Post-occupancy Evaluation**, review of **Project Performance**, **Project Outcomes** and **Research and Development** aspects. Updating of **Project Information**, as required, in response to ongoing client **Feedback** until the end of the building's life.
Sustainability Checkpoints	**Sustainability Checkpoint - 0**	**Sustainability Checkpoint - 1**	**Sustainability Checkpoint - 2**	**Sustainability Checkpoint - 3**	**Sustainability Checkpoint - 4**	**Sustainability Checkpoint - 5**	**Sustainability Checkpoint - 6**	**Sustainability Checkpoint - 7**
Information Exchanges (at stage completion)	**Strategic Brief**.	**Initial Project Brief**.	Concept Design including outline structural and building services design, associated **Project Strategies**, preliminary **Cost Information** and **Final Project**	Developed Design, including the coordinated architectural, structural and building services design and updated **Cost Information**.	Completed Technical Design of the project.	**'As Constructed' Information**.	Updated **'As Constructed' Information**.	**'As Constructed' Information** updated in response to ongoing client Feedback and maintenance or operational developments.

Project Roles Table

	0 Strategic Definition	1 Preparation and Brief	2 Concept Design	3 Developed Design	4 Technical Design	5 Construction	6 Handover and Close Out	7 In Use
Client	Client - Oldsworth							
Project lead	NDG Architects	NDG Architects	NDG Architects	NDG Architects	NDG Architects	NDG Architects	NDG Architects	NDG Architects
Lead designer	NDG Architects	NDG Architects	NDG Architects	NDG Architects	NDG Architects	NDG Architects	NDG Architects	NDG Architects
Construction lead					Contractor	Contractor	Contractor	Contractor
Architect	NDG Architects	NDG Architects	NDG Architects	NDG Architects	NDG Architects	NDG Architects	NDG Architects	NDG Architects
Civil and structural engineer			Structural engineer	Structural engineer	Structural engineer	Structural engineer		
Building services engineer			Buildign services engineer	Buildign services engineer	Buildign services engineer	Buildign services engineer		
Cost consultant	quantity surveyor	quantity surveyor	quantity surveyor	quantity surveyor	quantity surveyor	quantity surveyor	quantity surveyor	
Contract administrator					NDG Architects	NDG Architects	NDG Architects	
BREEAM	Sustainability consultant BREEAM	Sustainability consultant BREEAM	Sustainability consultant BREEAM	Sustainability consultant BREEAM	Sustainability consultant BREEAM	Sustainability consultant BREEAM	Sustainability consultant BREEAM	Sustainability consultant BREEAM
Interior designer			Interior designer	Interior designer	Interior designer	Interior designer	Interior designer	
Landscape architect			landscape designer	landscape designer	landscape designer	landscape designer	landscape designer	
Lighting designer			Lighting designer	Lighting designer	Lighting designer	Lighting designer	Lighting designer	
planning consultant			Planning consultant	Planning consultant	Planning consultant			

Contractual Tree - Olsworth barn - Traditional Building Procurement method



Notes

This chart illustrates the contractual relationships that the client will have with the various project consultants during the procurement process

Multidisciplinary Schedule of Services

0 - Strategic Definition

Project role	Party	Tasks to be undertaken
All roles		
Client	Mr. & Mrs Oldsworth	Provide preliminary list of requirements - Client "wish list"
Project lead	NDG Architects	Arrange preliminary meeting with client to discuss requirements
		Clarify project team roles and responsibilities
		Establish Outline Project Programme
		Provide Feedback from previous projects and experience.
Lead designer	NDG Architects	Contribute to preparation of Strategic Brief - prepare report for client review
		Comment on project Programme
		Provide Feedback from previous projects within the office
Architect	NDG Architects	Items to considerand/ or research that will have impact on the brief: Any protected Trees; any protected Wildlife; Flood risk; Condition of existing building; Contaminants in soil if barn was used for storing rural equipment; connections to sewers and municipal services such as water and sewage and power; any way leaves; any easements; site access for workers.
		Review local authority planning framework with special focus on location of project in National park. Research if any part of the project is Listed by English Heritage
		Provide Feedback from previous projects and in consultation with regards to schedule of required statutory approvals - Planning, English Hertitage, National Park Authority, Building Control
Building services engineer	TBD	Contribute to preparation of Strategic Brief in terms of engineering stategies particularly those incorporating sustainable strategies
Civil & structural engineer	TBD	Contribute to preparation of Strategic Brief referencing necessary studiesand investigations of existing building and surrounding soil conditions
Cost consultant	quantity surveyor	Contribute Cost Information to preparation of Strategic Brief using similar projects as a guide
Sustainability consultant	BREEAM consultant	Provide advice on sustainability strategies that may be adopted. Review sustainabilty goals of client and comment on possible environmental impact of the project

Multidisciplinary Schedule of Services

1 - Preparation & Brief

Project role	Party	Tasks to be undertaken
All roles		Provide information for and contribute to contents of Project Execution Plan as required
Client	0	Contribute to development of Initial Project Brief including Project Objectives, Quality Objectives, Project Outcomes, Sustainability Aspirations, Project Budget and other parameters or constraints
Project lead/ Lead designer	NDG Architects	Develop Initial Project Brief with project team including Project Objectives, Quality Objectives, Project Outcomes, Sustainability Aspirations, Project Budget and other parameters or constraints
		Collate comments and facilitate workshops as required to develop Initial Project Brief
		Prepare Project Roles Table and Contractual Tree and continue assembling and appointing project team members
		Prepare Schedule of Services and develop Design Responsibility Matrix including Information Exchanges
		Review Project Programme and Feasibility Studies
		Prepare Handover Strategy, Risk Assessments and Project Execution Plan
		Monitor and review progress and performance of project team
Architect	NDG Architects	Contribute to preparation of Initial Project Brief
		Discuss project with appropriate planning authority
		Undertake Feasibility Studies
		Prepare Site Information report
Building services engineer	TBD	Contribute to preparation of Initial Project Brief
		Contribute to Site Information report
Civil & structural engineer	TBD	Contribute to preparation of Initial Project Brief
		Contribute to Site Information report
Cost consultant	quantity surveyor	Contribute to preparation of Initial Project Brief
		Prepare Project Budget in consultation with client, architect, project lead and other consultants as required
Sustainability Consultant	BREEAM Consultant	Cooment and contribute to Initial project brief. Attend briefing meeting with client to answer questions regarding sustainable strategies

QUESTION 8

09.4.2014

To: micahelhart@ndgarchitects.com

From: candidate@ndgarchitects.com

CC:

Re: Listed Library in Madchester

Advice to action group regarding Listed building alterations

Dear Michael,

In reference to your email today, Please find below points that you can use in your meeting with the Action Group next week.

A We can review the listed status of the building on the English Heritage website. The listing will give varying degrees of detailed information about the building, hopefully there will be some information with regards to what elements are listed. We can set up a meeting with English Heritage on site to review the building. The Conservation officer will be able to give us clear information on which particular parts of the structure are listed if the Listing entry was not detailed enough. In addition he will be able to give us advice as to how we may go about designing the proposed facilities to fit within the building structure in a way that is sensitive to the building's listed status.

You may wish to point out to the Action Team that English Heritage understands that although buildings are Listed for a variety of reasons, if buildings are not able to adapt to new uses in the future then they will lose their cultural value and usefulness to society. Thus English heritage can actually be quite flexible in terms of the adaptation of buildings so that they retain their usefulness without losing their listed character.

B In designing the new toilet and kitchenette we will have to take into account the recommendations given to us by the English Heritage case officer. I expect that he will recommended that we look for ways to run new services that will not disrupt the listed elements of the building - it sounds like an under used store room at the rear of the building will not contain important historical features so I believe this sounds like a good choice of location. We will have to make a planning permission submission and concurrently an application to English Heritage once our designs are complete at RIBA stage 3.

We will also have to submit an application for Building Control approval under Building Regulation G – Sanitation, hot water safety and water efficiency. As the toilet is in a public building and there does not seem to be one in the building already we will also have to address issues of Inclusive Design and handicapped accessibility in compliance with Building regulation M

C Process and documentation required:

- Pre- application to LPA – No change of Use Class but extent of the proposed works means Planning permission and listed building consent will have to be obtained concurrently. Simple preliminary sketches will suffice for a pre-application which will be reviewed by the panning office who may request a meeting to review and give feedback for the full application. Meeting with English heritage has been discussed above.
- Design Drawings and details (Stage 3) for both application (Planning Permission and Listed building) will be submitted over the Planning portal online. There will need to be an application from filled including Certificates of Ownership, the fee (there is not fee however for listed building applications), Design and Access statement, Surveys and any other required reports such as any protected wildlife that is on the site.
- The application is received and checked
- If everything is in order with the application it is Registered.
- The project will either go to planning committee or be decided by delegation
- Permission, if granted, will normally come with conditions. Permission is not granted until conditions are met
- Full plans application with relevant fee for Building regulations approval will be made well in advance of work starting on site. Full plans application is required as I assume that the works will take place within 3 meters of an existing drain or sewer. Sufficient plans and details to show intended construction following requirements of the relevant regulations

D Timescale:

- Planning Permission and Listed building consent within 8 weeks as this is not a "major project" Within this period is a phase of public consultation which is 3 weeks.
- Once granted, permission is granted for 3 years.
- Local Authority will liaise and consult with the relevant authorities (fire and sewerage)
- Decision notice will be issued within 5 weeks.
- Full Plans approval Notice is valid for 3 years

E The repairs to the lead flashings will also require listed building consent as they sound extensive and as they were stolen off the building are not part of routine maintenance.

I trust the above is useful. Please let me know if you need anything else

Best regards

Candidate

NDG ARCHITECTS
Wharf Mill Paccadilly, Madchester, MAD PPA

EXAMINATION QUESTIONS

QUESTION 9

SECTION 05

09.4.2014

To: BelindaCarter@ndgarchitects.com

From: candidate@ndgarchitects.com

CC:

Re: Refurbishment and Conversion of existing stables into Farm Shop and café at Tally Ho Manor

Project staffing issue

Dear Belinda,

Great news that this project is finally moving ahead! I am certainly aware of our present workload and this week in particular has been especially intense. With regards to your suggestion to perhaps employ a Part 1 Architectural Assistant under the supervision of an Associate I would offer the following thoughts:

- The project will be moving into RIBA work stages 4, 5 and 6. These stages are crucial from the point of view of accurate and well-coordinated production and technical design information.
- While the participation and supervision by an associate is what I think these stages definitely require, I think that there will be too much of a "knowledge gap" between an associate and a Part 1 assistant to create an effective team. The associate may have to spent too much time training and instructing the assistant with possible detrimental affect to the project programme
- During Stage 5 inspections of the site will have to be made by our firm, it will probably be beyond the capabilities and experience of a Part 1 assistant to effectively analyse what he or she is seeing on site in order to make the site visits effective within our scope of work
- Lack of experience on job site, particularly where existing buildings are concerned can be unsafe for an inexperienced site visitor. An associate may not have enough time to accompany the Part 1 assistant on every site visit and putting employees at risk unsupervised on a job site is definitely a bad idea for all concerned.
- Taking on a full time employee, even at a lower salary level, brings with it a host of overhead costs that need to be factored into a practice's cash flow.
- I suggest that we should consider that a more senior and experienced post Part 3 Architect would be better choice for this project.
- A qualified architect will have the right balance of professionalism, competence and skill to be more self sufficient, freeing up the practice associate to work on other projects while still performing a more normal supervisory role in this refurbishment project.
- Options for this new employee would be to hire a Temporary Employee for a limited period, for instance, until the project completion. In this way salaries and overheads can be funded by fees

generated by the project being on site and when the project is over, the salary and overhead ceases too.

- Alternatively, though this can be a more expensive option, we could hire and architect from an Agency. Advantages are that the architect can be available to start immediately and the recruitment agency is responsible for all benefits etc. Having this employee be an "at will" hire gives you the flexibility to lay them off without complications should the project come to a halt again.
- Disadvantages of the temporary or agency employee staffing solution is the staff member not having a vested interest in the overall development of the practice which can lead to a lack of cohesiveness in the organic "make up" of the personnel in the office. You mention that we have not hired new staff for some time – anyone new, even if temporary, can end up shifting the natural balance of staff that has developed over time. However, sometimes this can be a good thing too!

Finding the right person:

The type of staff suggested above can be recruited either directly through an Employment Agency (where we agree the terms of the hire directly with the agency – fastest way to get new staff), or through a Specialist Recruitment Agency that would supply resumes of likely candidates for interview. Alternatively, for the temporary employee route, we can advertise in the trade press online, (RIBA, AJ and BD) with an advertisement for Temporary employee for a specific project. In addition, it is always worthwhile to have a "word of mouth" approach to recruiting staff running in parallel. Often times, suitable candidates to work on contract basis are available as they are in between jobs or finishing some other contract up. As you know, architects are often friends with other architects and "putting the word around" about a temporary position can be an effective recruiting tool.

Below is draft suggestion for a job description/ advertisement suitable for recruitment either through the press or a Recruitment Agency:

> Medium Sized Practice in the North West UK is looking for a post Part 3 qualified Architect to work on a unique refurbishment and conversion project on a historic rural property. The project is currently entering RIBA Stage 4
>
> Good CAD skills required as well as effective project running skills
>
> At least 1 year post Part 3 experience required as well as experience in production of coordinated working drawings and on-site operations. Experience of refurbishment of historical structures and small scale hospitality/retail projects would be considered advantageous
>
> Temporary position with salary of £2,500 per month. Benefits to be determined with individual candidate
>
> Must be self-motivated and have good team working skills
>
> *Practice is an equal opportunities employer*

Below is a draft interview checklist:

- Establish if there will be single or multiple interview
- Who will in charge of interview process? I suggest a member of senior management staff and the Associate that will be supervising the project.
- Check to see if any work permits are required by the candidate
- Does the candidate have the skills, expertise and experience to perform the job?
- Is the candidate enthusiastic and interested in the job and the company?
- Will the candidate fit into the team, culture and company?
- Possible questions to ask of the candidate at the interview:
    - "What interests you about this job, and what skills and strengths can you bring to it?"
    - "Can you describe a professional achievement that you are proud of?"
    - "What sort of work environment do you prefer? What brings out your best performance?"
    - "Why did you leave your last job, and what have you been doing since then?

Contractual arrangements will depend very much on the nature of the hire. If the hire is through an employment agency than the contract will be between our firm and the agency. In the case of a temporary or contract employee then the terms of the contract should include;

- Payment system / tax / National Insurance arrangements
- Systems of complaints / disciplinary processes
- Probation period / Notice for termination
- Staff review systems, appraisals, maternity, paternity, sickness, holidays, CPD, working hours, data protection
- Staff pension scheme and share holder schemes will probably not be relevant to an employment contract this type

I hope that the above proves useful to you making any final decisions as to how we will staff this project.

Best regards,

Candidate

NDG ARCHITECTS
Wharf Mill Paccadilly, Madchester, MAD PPA

EXAMINATION QUESTIONS

QUESTION 10

SECTION 05

09.4.2014

To: JamesNoble@ndgarchitects.com

From: candidate@ndgarchitects.com

CC:

Re: City Gallery Extension

Project Procurement Options

Dear James,

This sounds like a very exciting and prestigious project. I am happy to help formulate some constructive thoughts on procurement options and how you might make some suggestions at your next meeting.

Firstly, we can consider the project framework:

This is a quite large project in an existing building. It is complicated in terms of access, site constraints, the necessity to retain public access during construction and seeing as you are proposing that the extension be on the roof, it will probably utilize innovative design and advanced engineering.

Where a project is of a certain scale, is complicated and requiring innovation there is likely to be an increase in the complexity of the procurement. There will likely be many interested parties - funders, clients, specialist contractors, suppliers and consultants for innovative structures and roofing techniques, engineers, lighting designers as well as cost consultants, graphic designers and operations managers. In light of these multiple and sometimes competing interests it is likely that we will need to choose a sophisticated procurement method as you suggest.

Secondly, we can consider the key players in the project:

Sir Jack Draw – archive director:

- You and our firm have a good and open relationship with him.
- He is open minded about the project design.
- He is not averse to certain amount of project risk – he has a certain level of experience in construction projects and an apparent "a hands" on approach
- Time and design quality seem to be at the top of his agenda.

- I expect his apparent willingness to carry more risk is to achieve the best possible price with the fastest possible procurement route
- We will need to review with Sir Jack who are the funders of this project and where the funds will be coming from – bank loans, donors or private sources
- We will need to review the liquidity of the funding as well as the level of risk that the funders are prepared to accept as it will have an impact on the chosen procurement route
- We will need to assess whether his organisation has the resources to manage the project or if a third party will need to be appointed to perform the management function on his behalf.

Chief Planning Office – Local planning Authority:

- We have good and open relationship with him.
- He has a positive attitude about the proposed scheme in outline terms – it was just a conversation, but a positive one!

NDG Architects – James Noble, Director:

- You have good personal relationships with the other key stakeholders.
- You are looking to demonstrate an innovative approach to design and construction.
- As an architectural practice owner you are comfortable with risk to varying degrees!
- You have been scaling back your involvement in projects in the last year or so and you just informed us all that you will be retiring with immediate effect.

Thirdly, we can consider the possible procurement framework;

We are looking to find the right balance between the, oft quoted, project goals of high quality, high speed and low price. Time is a priority. We don't know if the client is prepared to pay more for a faster completed project. Some procurement methods are better suited to an earlier start on site and therefore an earlier finish. We need to determine whether and the extent to which the design should be allowed to change during the construction phase – we will be working in an existing building and could encounter unforeseen features that would require the design to change. We need to determine the importance to the project funders of obtaining a fixed price for the works. With regards to management of the procurement process as whole, it may be advantageous to consider the employment of a third party project manager. This would not be an appointment to replace the traditional architects roles who would remain as lead consultant. This project manager could take on the role of contract administrator and be named as such in the contract documents. The advantage of this position, provided the right person is selected, is one central management position for the entire team to work with, managing the relevant stakeholders of the entire process and driving the process forward. I believe that this role would enable you to fulfil your intention to act as consultant to the firm going forward while either your co-director or one of your senior directors could take on day to day responsibilities with regards to this prestigious project.

And the available procurement options:

Traditional: JCT SBC with Quantities

JCT Intermediate Building contract

Advantages: completed design leads to a fixed cost. Straightforward contractual relationships that work for experienced and inexperienced clients and form projects of varied complexity

Disadvantages: Work will not start on site until design is finalised leading to potential increase in time from inception of project. Greater possibility for an adversarial relationship between the design team and consultant team. Responsibilities shared amongst project team

Design Build: JCT Design Build Contract

Advantages: Cost conscious route. Client born risks are effectively transferred to the contractor – contractor becomes client's single point of responsibility

Disadvantages: Design quality tends not to be high priority. Lack of control of design elements could lead to issues with planning applications and other statutory bodies if the design built team does not include original design architect

Two stage tendering: JCT SBC with Approx. Quantities

Advantages: Design process can overlap with construction works. Contractor is involved in project early and can advise on cost and "buildability" issues

Disadvantages: Given that tenderers are narrowed down at second stage there is less competition with regards to getting best price

Construction management: JCT Construction Management appointment

Advantages: High speed – design and works can move forward in parallel. Employer in control of the process – promotes teamwork

Disadvantages: uncertainty with regards to time for completion and price. Responsibility is spread around entire team but client retains responsibility for performance of design team

Management contracting: JCT Management Building Contract

NEC3: Engineering & Construction Contract

Advantages: High speed – design an works can move forward in parallel with the added advantage that a timely completion is also possible due to centralised management framework as well as financial incentives for the management contractor. Scope for client changes

Disadvantages: uncertainty with regards to price. Although preliminaries and management fee can be fixed, price certainty only comes at later stages of construction process – prime cost. Tough discipline on all sides – early design decisions have "knock on" effects – design team will need strong and constant leadership. Only as good as the quality of the management contractor and his relevant experience.

In my opinion and at this early stage of the project with limited knowledge, a choice from last three options would give us the best procurement route for the type of project. We need clearer answers to some of my earlier questions in order to make a final selection of the three best options, but hopefully you will have found my notes useful in preparing for your next meeting

Best regards,

Candidate

NDG ARCHITECTS
Wharf Mill Paccadilly, Madchester, MAD PPA